# Teach Chhota 📚

5-Minute Judge Pitch Presentation Script & Defense Dossier

⏱️ **STRICT RUNTIME: 4m 45s**

Target: 620 words @ 135 WPM | Max Cap: 5:00

| Timestamp | Section | Core Message / Judge Takeaway | Pacing |
|---|---|---|---|
| **0:00 – 0:40** | 1. The Hook & The Problem | Passive learning yields 70% loss; EdTech ignores tier-2/3 offline reality | 40s (88 words) |
| **0:40 – 1:20** | 2. The Innovation & Solution | The Protégé Effect & Feynman Technique: Student teaches an AI tutee | 40s (92 words) |
| **1:20 – 2:50** | 3. Live MVP Demonstration | Teach Chhota studio, Galti-Dex, Knowledge Tree & Parent Hindi Audio | 90s (195 words + demo) |
| **2:50 – 3:40** | 4. Tech Stack & AI Engine | Gemini 2.5 Flash RAG, Samajh Engine, SM-2 Spaced Repetition, Offline PWA | 50s (115 words) |
| **3:40 – 4:20** | 5. USPs & Defensibility | Collectible mistakes (Galti-Dex), No-Anxiety metric, True Offline-First | 40s (85 words) |
| **4:20 – 4:45** | 6. Business Viability & Vision | B2C Freemium, B2G DIETs heatmaps, Bharat accessibility | 25s (55 words) |
| **4:45 – 5:00** | 7. Buffer / Transition | Smooth handoff to Judges' Q&A session | 15s buffer |

## Act 1: The Hook & The Broken EdTech Paradigm

**0:00 – 0:40 (40 Seconds)**

> ***Visual:*** *Slide 1 (Split graphic: Child passive before video vs. Child enthusiastically explaining). Stand center stage. Deliver with clear, urgent conviction.*

"Respected judges, here is the harsh reality of Indian EdTech: Millions of students watch high-definition video lectures every day, yet their conceptual retention drops by over **70% within 48 hours**. Why? Because **watching a video is not learning**.

Worse, 95% of EdTech tools assume high-speed 5G. For over **150 million school students across Bharat**, expensive subscriptions and spotty connections lock them out.

We asked: What if, instead of another AI bot lecturing the child, we flipped the classroom? **Meet Teach Chhota.**"

## Act 2: The Core Innovation — The Protégé Effect

**0:40 – 1:20 (40 Seconds)**

> ***Visual:*** *Slide 2 (Mascot Chhota with speech bubble: "I get confused easily! Can you teach me?").*

"Teach Chhota is an offline-first, AI learning companion for Classes 6 to 12. But Chhota is **not a tutor**.

Chhota is a friendly, curious AI student who holds common academic misconceptions. **The child is the teacher**.

Grounded in cognitive science's **Protégé Effect** and the **Feynman Technique**, students master Math and Science by explaining concepts aloud to fix Chhota's wrong beliefs. Because when you teach it, your brain permanently keeps it."

## Act 3: Live MVP Product Walkthrough

**1:20 – 2:50 (90 Seconds)**

> ***Visual:*** *Live Screen Demo of Teach Chhota web app. Seamless transitions between tabs.*

**[1. Practice Studio – /teach]:**
"Here in our Practice Studio, Chhota states a classic doubt: '*I think 1/8 is bigger than 1/4 because 8 is bigger than 4!*'
The student taps 'Hold to Talk' and explains: '*No Chhota, think of cutting a pizza into 8 slices versus 4. 8 slices are much smaller!*'
In real time, our AI evaluates correctness, clarity, and analogy use, and Chhota celebrates: '*Oh, I get it now! Larger denominators mean smaller slices!*'

**[2. Galti-Dex – /galti-dex]:**
"Next, notice what happens: Debunking that misconception unlocks a collectible card in the **Galti-Dex**. We turn the shame of mistakes into a Pokédex-style trophy hunt with Daily Bug Bounties!

**[3. Knowledge Tree & Parent Audio – /tree & /parent]:**
"Concepts live on a **Knowledge Tree** that wilts if not reviewed, triggering spaced repetition. And for parents, we generate zero-anxiety audio summaries in Hindi and English with zero marks and zero stress."

## Act 4: Technical Architecture & AI Stack

**2:50 – 3:40 (50 Seconds)**

> ***Visual:*** *Slide 4 (Architecture flow: React PWA ⇄ Express Server ⇄ Gemini 2.5 Flash / Embeddings ⇄ Local SQLite).*

"Our engineering stack is built for scale, intelligence, and edge reliability:

• **AI & RAG Pipeline:** Powered by **Google Gemini**. We use `gemini-embedding-001` to index NCERT textbooks, and `gemini-2.0-flash / 2.5-flash` for syllabus-grounded doubt resolution and multi-factor evaluation.
• **The Samajh Score Engine:** A rigorous mathematical formula weighting correctness (35%), misconception resolution (25%), clarity (25%), and analogy use (15%), multiplied by an **SM-2 spaced repetition decay** that rewards long-term memory over cramming.
• **Offline-First Resilience:** Built with **React 18, Vite, TypeScript, Drizzle ORM, and SQLite**. If the internet drops, local keyword fallbacks and browser Web Speech APIs ensure the student never stops learning."

## Act 5: Unique Selling Propositions (USPs) & Moat

**3:40 – 4:20 (40 Seconds)**

> ***Visual:*** *Slide 5 (Competitive Comparison Grid).*

"Why can't Byju's or ChatGPT easily copy this?
1. **Flipped Role (Tutee vs Tutor):** While ChatGPT answers homework and promotes cognitive laziness, Teach Chhota forces cognitive effort through the Protégé Effect.
2. **Galti-Dex Gamification:** We destigmatize failure by turning errors into collectible assets.
3. **Bharat-Ready Architecture:** Zero-latency client-side speech, Hindi audio readouts, and full offline survivability designed for budget smartphones."

## Act 6 & 7: Business Viability, Social Impact & Grand Close

**4:20 – 4:45 (25 Seconds) + Buffer**

> ***Visual:*** *Slide 6 (Mascot Chhota smiling, QR code to live demo, team credits). Confident closing tone.*

"Our business model combines a **Freemium 'Teach Chhota Plus'** tier with **B2G school partnerships** providing non-ranking misconception heatmaps to teachers.

Judges, the best way to master anything is not to be lectured by a supercomputer—it is to have the confidence to teach someone else. With Teach Chhota, we empower 150 million young learners to become masters of their own mind.

**Thank you! We welcome your questions.**"

| Feature | Generic Video Apps | ChatGPT / Homework Bots | Teach Chhota 🌟 |
|---|---|---|---|
| **Learning Dynamic** | Passive viewing (<20% recall) | Passive copying (Zero recall) | **Active Teaching (>80% recall)** |
| **Treatment of Errors** | Penalized on quizzes | Ignored (Gives direct answers) | **Celebrated in Galti-Dex** |
| **Connectivity Need** | High-bandwidth video streaming | Constant internet required | **Offline-First PWA + Local RAG** |
| **Parental Anxiety** | High (Scores & rankings) | None / Detached | **Zero-stress Hindi/English audio updates** |

## 🎯 Judge Q&A Defense Strategy (Quick-Fire Answers)

**Q1: "How can it be truly offline-first if you are using Google Gemini?"**

**Answer:** "Teach Chhota uses an *Offline-First, Graceful Degradation* design. When connected, queries tap Gemini 2.5 Flash for nuanced evaluation and embeddings. When offline, our Service Worker caches core NCERT passages, runs local heuristic similarity scoring, and leverages native on-device browser speech APIs. The core learning loop never breaks."

**Q2: "What if the student hallucinates or teaches Chhota something completely wrong?"**

**Answer:** "Chhota does not blindly accept inputs. Every explanation is checked against our NCERT ground-truth knowledge base. If the student reinforces the mistake, Chhota expresses gentle confusion: *'Wait, does having 8 parts really mean bigger slices?'* and prompts them to try again with a scaffolded hint."

**Q3: "How do you stop students from gaming the system to hoard coins?"**

**Answer:** "Our *Samajh Engine* applies mathematical diminishing returns: repeating the same concept lowers score gains by up to 85%. Furthermore, our SM-2 algorithm enforces a 0.7x penalty on cramming, but rewards a 1.5x multiplier when reviving wilted concepts after the proper memory consolidation window."